iNavIC (both by India). iNavIC is the regional satellite navigation system of India launched and operated by Indian Space Research Organisation (ISRO). It covers India and nearby regions extending up to 1500 kilometres.

Satellites are spinning objects in an orbit in space that can tell you where you are exactly. You just need to have a GPS receiver that can pinpoint your position, to just a few metres. The GPS receiver can guide you on your way to your destination, also powered by a voice that will direct you to go left, right or straight by finding a proper route to your destination. It is a broadcast system based on radio signals that reach all parts of the Earth, and can be used by any number of people at once.

Apart from helping people reach their destination, satellite navigation can also help track parcels, stolen properties, vehicles and growing crops. It helps find missing persons, guide the blind, improve traffic flow and vehicle efficiency, and conduct scientific research in meteorology, troposphere and geodesy. Despite the extensive importance of GPS in everyday life, most of us do not know how it works. So, let us take a closer look.

How satellites help in navigation

All satellites are constantly beaming out radio-wave signals towards Earth from a known position, and these signals travel at the speed of light. Every signal contains information about the satellite it came from and a time stamp that informs when it left the satellite. By noting when each signal arrives, the receiver can figure out how long it took to travel and how far it has come. Each satellite carries extremely precise atomic clocks to calculate how long the signal takes to travel. If a receiver picks up a signal from three or four different satellites, it can figure out your precise location, including altitude.

GPS has three major segments:

- Space segment, which consists of a network of satellites.
- Control segment, which is a ground control network of antennae, monitors and control stations that manage the satellites, and the receiving device you carry with you.
- User segment, which is an electronic receiver you hold in your hand or carry in your vehicle.

Radio waves travel at the speed of light but only in a vacuum. These signals beaming down to us from space satellites travel through Earth's atmosphere, including the troposphere, the uncharged region of the atmosphere and the ionosphere that contains charged particles. The troposphere and ionosphere distort and delay satellite signals in quite complex ways. So, GPS receivers have to compensate to ensure these can make accurate measurements of distance. The difference between the time it takes for the signal to be emitted by the satellite and for the satellite to receive a signal back from the receiver is tightly calculated and used to determine exactly where the receiver is located.

Yellow, blue and red spheres are the coverage areas of three different satellites, as shown in Fig. 1. If the receiver picks up a signal from these three satellites, it must lie somewhere at the black dot. Similarly, any object receiving a signal from any of the three satellites can be located from their meeting point. The signal from the fourth satellite will help find the altitude as well. Since there are many more GPS satellites, you can locate yourself wherever on Earth you happen to be.

Satellite-based navigation in ships

Navigation in ice-covered waters is achieved through images from radar satellites. Ice maps generated from satellite data are sent to the ships in addition to radar images. These maps make it possible to draw conclusions about ice thickness and navigability. This helps rescue teams better evaluate the immediate ice situation and avoid unnecessary detours, saving fuel. It also helps in the mapping and measuring of oceans, tides, icebergs and sea levels.

This ensures safe shipping by protecting seas and coastal waters, and assisting public authorities to combat unlawful activities such as hazardous substance dumping, illegal fishing and piracy.

Fig. 1: Three spheres representing the coverage areas of three different satellites (Credit: www.explainthatstuff.com)

New processors have been developed that can generate ice maps directly after acquisition of satellite images. These can determine the location and extent of various types of ice. To do this, complex data comprising image features that escape the human eye are analysed. Other processors are being developed to pave the way for detailed insights into the movement of sea ice in high-resolution images of ice floes.

Synthetic aperture radar (SAR). This technology can generate images from a moving platform, regardless of light and clouds. A SAR sensor is an active system that carries an illuminator with it to sense in the dark. It uses microwaves that have wavelengths much larger than visible light. Because of that, SAR signals can penetrate through clouds and semi-transparent materials like ice, snow, vegetation, etc.

There are different types of SAR. If the radar transmits and/or receives in two different polarisations—horizontal and vertical—it could help retrieve false-colour images instead of just black-and-white data. This is called polarimetry, and it can be used for precision farming or for water bodies monitoring in floods, among others.

Another option is to acquire images from slightly different angles, in a technique called interferometry, which provides 3D vision.